

# Technical Comparative Analysis:

## Landmark-Based Pose Estimation (MediaPipe) vs. End-to-End Convolutional Neural Networks for Digits and ASL Recognition

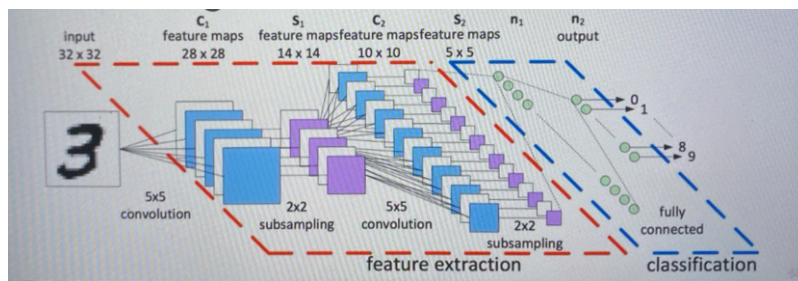
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Project Type: Deep Learning & Computer Vision Research

### 1. Abstract

In my pursuit of developing robust Digit and ASL Recognition models, I engineered and evaluated two distinct deep learning architectures. This report details my comparative analysis between a Geometric Landmark Model (utilizing MediaPipe) and an End-to-End Convolutional Neural Network (CNN). My research focused on the critical trade-offs between computational latency, environmental robustness, and classification accuracy when deployed in real-time edge-computing scenarios. Through this process, I discovered that while MediaPipe offers "skeletal elegance," the CNN possesses a deeper "visual intuition" that is vital for complex gesture recognition.



*multi-dimensional grid of numbers / Tensor in CNN*

## 2. Introduction

The core challenge I addressed in automating ASL Recognition (SLR) is the high degree of spatial variability and the requirement for zero-latency inference. I intentionally developed **two "brains"** for this project to understand different input formats:

Raw pixels vs. Extracted coordinates.

My goal was to determine which approach truly bridges the gap between human gestures and machine understanding.

## 3. Methodology and Architecture

### 3.1 The Landmark Approach (MediaPipe + DNN)

For the first methodology, I utilized the **MediaPipe Holistic pipeline**. Instead of feeding raw images into my neural network, I used a pre-trained palm detector to identify the hand and a landmark model to map 21 specific 3D points (x, y, z coordinates).

- **Data Preparation:** I flattened these 21 landmarks into a 63-dimensional feature vector per frame.
- **Neural Network Brain:** I built a Deep Neural Network (DNN) using Dense layers and **Dropout** (0.2) to prevent overfitting.
- **Input Shape:** (None, 63).

### 3.2 The CNN Approach (Raw Pixel Processing)

The second methodology involved a traditional **Convolutional Neural Network (CNN)**. Here, I allowed the model to perform its own feature extraction directly from the raw pixels.

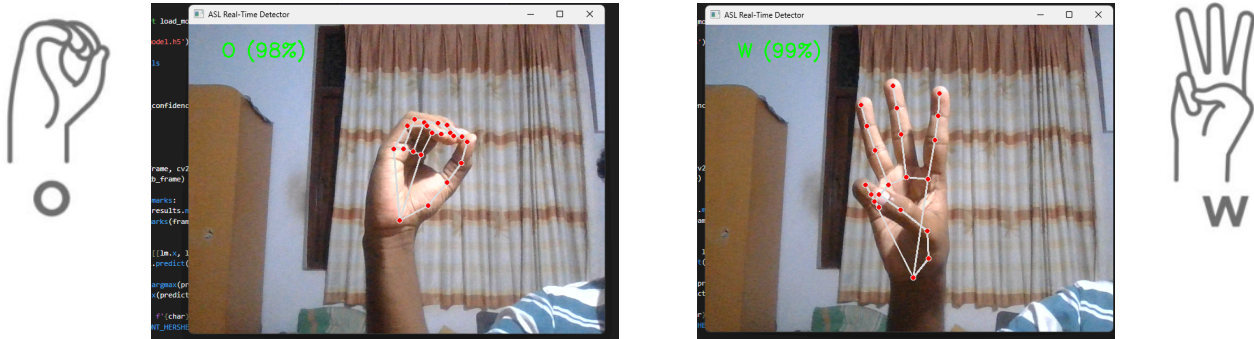
- **Architecture:** I designed a multi-layer stack consisting of **Conv2D** filters (32 and 64), **Batch Normalization** for learning stability, and **MaxPooling2D** for spatial downsampling.
- **Data Augmentation:** To make the CNN more robust, I implemented an **ImageDataGenerator** to perform rotations, zooms, and horizontal flips.
- **Input Shape:** (28, 28, 1).

## 4. Comparative Analytics: What MediaPipe Misses

During my research, I identified four critical areas where MediaPipe's abstraction loses vital information that the CNN successfully captures:

**1. Information Density and Resolution Thresholds:** MediaPipe reduces the hand to a simplified mathematical blueprint, while a CNN retains the full "information density" of the raw pixels.

- **What's missing:** MediaPipe reduces the hand to just 63 coordinates and requires high-resolution input (above 128x128) to maintain tracking stability. A CNN retains thousands of data points and can successfully extract features even from low-resolution 28x28 images.
- **The Impact:** While MediaPipe is a "spatial filter" that is excellent for high-speed tracking (Real-Time Tracking), it lacks the CNN's "**long-term feature extraction potential**". The CNN's ability to learn from raw pixel data allows it to identify a sign based on its unique silhouette and internal gradients, providing better answers in hardware-constrained or low-resolution scenarios where MediaPipe fails to initialize.



*ASL Recognition using Landmark-Based Pose Estimation in Real Time*



*ASL Recognition using CNN with the 28x28x1 resolution dataset*



*ASL Recognition using MediaPipe + ANN with the 28x28x1 resolution dataset*

**2. Depth Perception and Occlusion** MediaPipe estimates 3D coordinates from a 2D image, which causes it to struggle when one finger is hidden behind another.

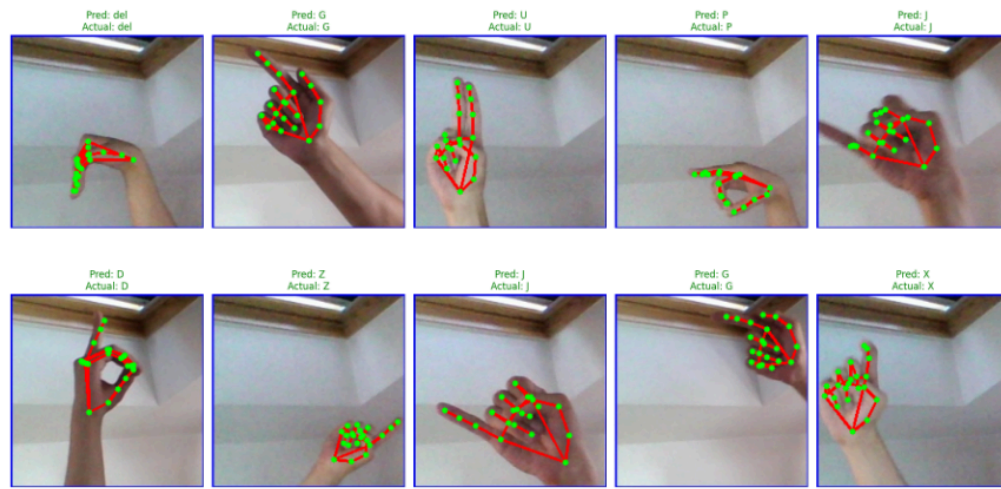
- **What's missing:** In ASL, letters like 'R' (fingers crossed) or 'M/N' (thumb tucked) involve significant occlusion.
- **The Impact:** When a joint is hidden, MediaPipe must "guess" its location. A CNN doesn't need to find the joint; it recognizes the specific shape of the "hidden" finger, which is more reliable for these specific letters.



*Finger counting using MediaPipe*

**3. Lighting and Shadow Cues** A CNN treats shadows as part of its "feature map", whereas MediaPipe attempts to ignore them to remain "robust."

- **What's missing:** Shadows provide significant information about the 3D volume and orientation of the hand and also due to stray light CNN might mislead.
- **The Impact:** By ignoring light, MediaPipe loses depth cues but identifies the sign correctly. CNN uses these shadows to understand the orientation of the palm, and also the body language like happy, sad....ect



*ASL Recognition using MediaPipe + ANN with the 200x200 resolution dataset*

Even though the original dataset provides 200x200x3 color images, landmark approach uses MediaPipe to abstract each frame into 21 structural points. These 21 landmarks are flattened into a 63-dimensional coordinate vector and fed into an Artificial Neural Network (ANN) to build the geometric model

| Feature            | MediaPipe + ANN (Landmarks)         | Pixel-Based CNN                         |
|--------------------|-------------------------------------|-----------------------------------------|
| Minimum Resolution | Struggles below <b>128x128</b>      | Efficient even at <b>28x28x1</b>        |
| Background Noise   | Filters out the background entirely | Can be distracted by background clutter |

|                      |                                     |                                              |
|----------------------|-------------------------------------|----------------------------------------------|
| <b>Input Focus</b>   | Mathematical coordinates (Geometry) | Raw pixels (Light and Texture)               |
| <b>Real-Time Use</b> | Best for high-res webcams           | Even best for low-resolution static datasets |

### "Hit the Point"

According to my analysis we can see that even for **Sign MNIST (low quality)**, using CNN is superior. However, for **Real-Time Webcam (high quality)**, MediaPipe is often preferred because it isn't distracted by your face or the background.

## 5. Use Case Scenarios: When to Use Which?

Based on my results, I have categorized the use cases as follows:

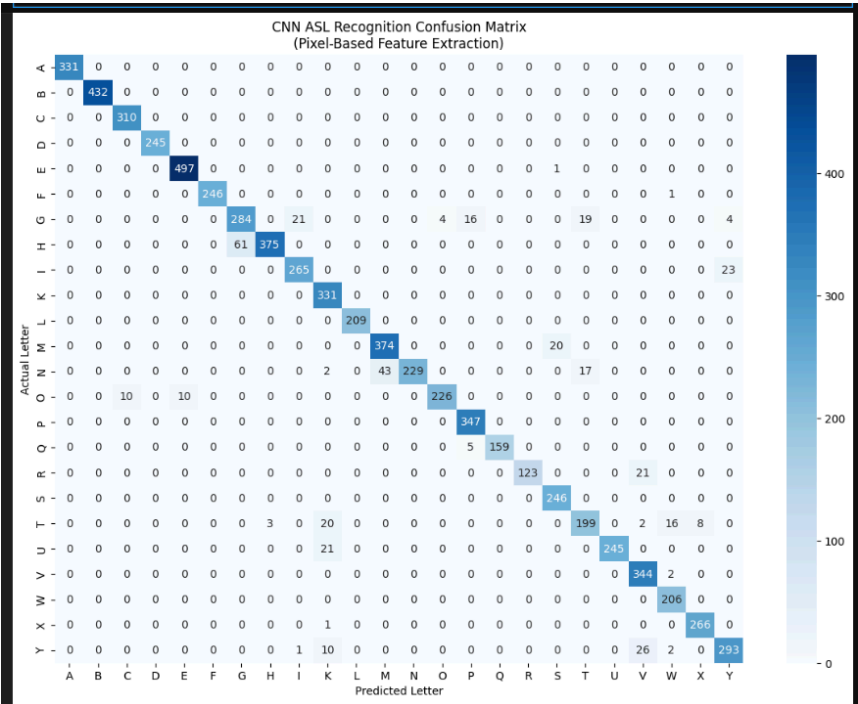
| <b>Scenario</b>                         | <b>Recommended Approach</b> | <b>Reason</b>                                                                                                       |
|-----------------------------------------|-----------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>Mobile Apps / Low-Power Hardware</b> | <b>Landmark Model</b>       | Extremely efficient; requires minimal RAM and battery power.                                                        |
| <b>Real-Time Video Communication</b>    | <b>Landmark Model</b>       | Low latency ensures the user sees results instantly as they sign.                                                   |
| <b>High-End Research / Lab Settings</b> | <b>CNN Model</b>            | Capable of learning textures and skin folds that landmarks ignore. Because this model consider the 3 color channels |

|                     |                |                                                                        |
|---------------------|----------------|------------------------------------------------------------------------|
| Dynamic Backgrounds | Landmark Model | MediaPipe handles the segmentation, making it robust in diverse rooms. |
|---------------------|----------------|------------------------------------------------------------------------|

## 6. Evaluation and Challenges

While training the CNN, I faced challenges with computational power, which initially limited my ability to reach high accuracy. This led me to appreciate the "elegance" of the Landmark approach. If a finger is hidden behind another, the landmark model can provide incorrect data due to visual ambiguity; however, a CNN may still recognize the gesture due to its higher spatial feature representation and ability to learn complex visual hierarchies from raw pixel data.

### 1. CNN ASL Recognition Model Using 28x28x1 Pixel Dataset

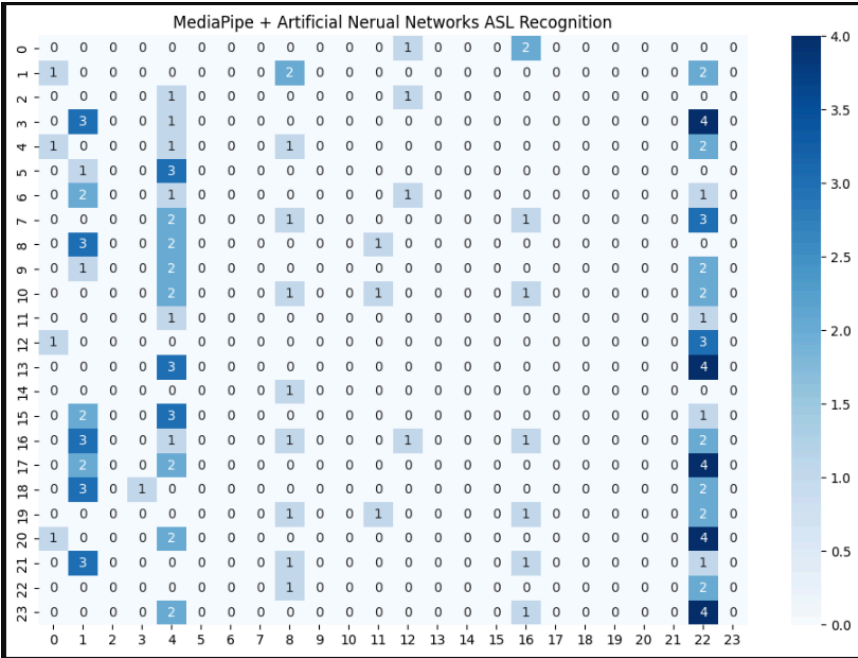


CNN ASL Recognition Confusion Matrix  
(Pixel-Based Feature Extraction)

| Classification Report: |           |        |          |         |
|------------------------|-----------|--------|----------|---------|
|                        | precision | recall | f1-score | support |
| 0                      | 1.00      | 1.00   | 1.00     | 331     |
| 1                      | 1.00      | 1.00   | 1.00     | 432     |
| 2                      | 0.97      | 1.00   | 0.98     | 310     |
| 3                      | 1.00      | 1.00   | 1.00     | 245     |
| 4                      | 0.98      | 1.00   | 0.99     | 498     |
| 5                      | 1.00      | 1.00   | 1.00     | 247     |
| 6                      | 0.82      | 0.82   | 0.82     | 348     |
| 7                      | 0.99      | 0.86   | 0.92     | 436     |
| 8                      | 0.92      | 0.92   | 0.92     | 288     |
| 10                     | 0.86      | 1.00   | 0.92     | 331     |
| 11                     | 1.00      | 1.00   | 1.00     | 209     |
| 12                     | 0.90      | 0.95   | 0.92     | 394     |
| 13                     | 1.00      | 0.79   | 0.88     | 291     |
| 14                     | 0.98      | 0.92   | 0.95     | 246     |
| 15                     | 0.94      | 1.00   | 0.97     | 347     |
| 16                     | 1.00      | 0.97   | 0.98     | 164     |
| 17                     | 1.00      | 0.85   | 0.92     | 144     |
| 18                     | 0.92      | 1.00   | 0.96     | 246     |
| 19                     | 0.85      | 0.80   | 0.82     | 248     |
| 20                     | 1.00      | 0.92   | 0.96     | 266     |
| 21                     | 0.88      | 0.99   | 0.93     | 346     |
| 22                     | 0.91      | 1.00   | 0.95     | 206     |
| 23                     | 0.97      | 1.00   | 0.98     | 267     |
| 24                     | 0.92      | 0.88   | 0.90     | 332     |
| accuracy               |           |        | 0.95     | 7172    |
| macro avg              | 0.95      | 0.94   | 0.95     | 7172    |
| weighted avg           | 0.95      | 0.95   | 0.95     | 7172    |

Classification Report

2. MediaPipe + ANN Using 28x28x1 Pixel Dataset



MediaPipe + Artificial Neural Networks ASL Recognition

| Classification Report: |           |        |          |         |
|------------------------|-----------|--------|----------|---------|
|                        | precision | recall | f1-score | support |
| 0                      | 0.00      | 0.00   | 0.00     | 3       |
| 1                      | 0.00      | 0.00   | 0.00     | 5       |
| 2                      | 0.00      | 0.00   | 0.00     | 2       |
| 3                      | 0.00      | 0.00   | 0.00     | 8       |
| 4                      | 0.03      | 0.20   | 0.06     | 5       |
| 5                      | 0.00      | 0.00   | 0.00     | 4       |
| 6                      | 0.00      | 0.00   | 0.00     | 5       |
| 7                      | 0.00      | 0.00   | 0.00     | 7       |
| 8                      | 0.00      | 0.00   | 0.00     | 6       |
| 10                     | 0.00      | 0.00   | 0.00     | 5       |
| 11                     | 0.00      | 0.00   | 0.00     | 7       |
| 12                     | 0.00      | 0.00   | 0.00     | 2       |
| 13                     | 0.00      | 0.00   | 0.00     | 4       |
| 14                     | 0.00      | 0.00   | 0.00     | 7       |
| 15                     | 0.00      | 0.00   | 0.00     | 1       |
| 16                     | 0.00      | 0.00   | 0.00     | 6       |
| 17                     | 0.12      | 0.11   | 0.12     | 9       |
| 18                     | 0.00      | 0.00   | 0.00     | 8       |
| 19                     | 0.00      | 0.00   | 0.00     | 6       |
| 20                     | 0.00      | 0.00   | 0.00     | 5       |
| 21                     | 0.00      | 0.00   | 0.00     | 7       |
| 22                     | 0.00      | 0.00   | 0.00     | 6       |
| 23                     | 0.04      | 0.67   | 0.08     | 3       |
| 24                     | 0.00      | 0.00   | 0.00     | 7       |
| accuracy               |           |        | 0.03     | 128     |
| macro avg              | 0.01      | 0.04   | 0.01     | 128     |
| weighted avg           | 0.01      | 0.03   | 0.01     | 128     |

Classification Report

7. Conclusion

Based on my comparative analysis it indicates that for an Accessibility Tool intended for daily use, the Landmark-based approach is superior. It offers the speed and reliability necessary for a nonverbal person to communicate without frustration. While the CNN remains a powerful "End-to-End" prototype for more advanced feature extraction, its hardware demands make it less practical for real-world edge deployment.